Simulation Analysis of the Japanese Economy with a Stock-Flow Consistent Model

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1. Research Content

○ Objective

To clarify the issues of the Japanese economy, which has been in a long deflationary situation, and to explore measures to place the economy on a sustainable growth trajectory.

○ Method Approach based on a macro model rooted in accounting (Stock-Flow Consistent Model)

⇨ Verified with actual Japanese SNA data (first attempt in Japan)

○ Key Points

- Currently, Japan is exploring achieving economic growth through wage increases.
- Meanwhile, the timing of interest rate hikes is being considered to address inflation and yen depreciation.
- While the economic effects of wage increases are generally recognized, the effects of interest rate hikes are divided (⇒ 2.)
- This study focuses on examining the effects of wage increases and interest rate hikes on the growth of the Japanese economy.

2. Effects of Interest Rate Hikes

○ Perspectives from Previous Research

・Keynesian and Neoclassical (Conventional)

Interest rates ↑ ⇨ Investment ↓ ⇨ GDP ↓ ⇨ Inflation rate ↓

・Post-Keynesian

Interest rates ↑ ⇨ Investment ↓ ⇨ Production capacity ↓ ⇨ Excess demand ⇨ Inflation rate ↑

・MMT

Interest rates ↑ ⇨ Asset income ↑ ⇨ Consumption ↑ ⇨ GDP ↑ ⇨ Inflation rate ↑

○ Results obtained from this research (Effects of interest rate hikes in Japan)

・OLS estimations : The correlation between corporate investment and interest rates is weak, or rising interest rates increase investment (⇒ 7.)

・Simulation result : Interest rate hikes stimulate economic growth through income effects (⇒ 9. 10.)

⇒ It is considered that the income increase effect outweighs the investment decrease effect (consistent with MMT)

3. Problem Awareness

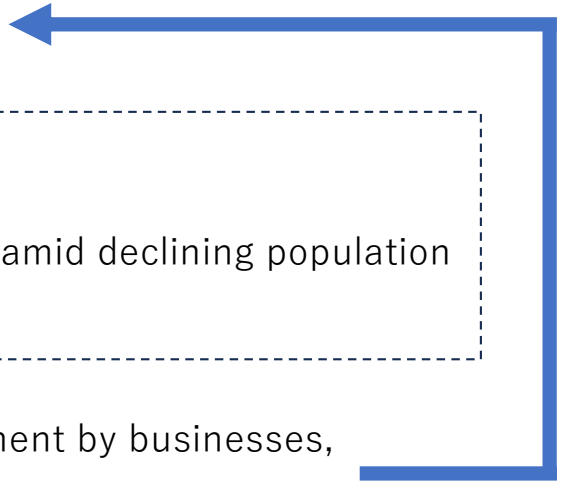
○ Stagnation of wages, prices, and the economy over 30 years

○ Concerns over the accumulation of government debt

○ Concerns about the sustainability of social security finances amid declining population and aging society

○ Low domestic consumption by households, domestic investment by businesses, and growth expenditure by the government

○ Unable to escape the deflationary spiral (the “Lost 30 years”)



4. Previous Research on SFC Models Referenced

○ Basic Texts on SFC Models

Godley, Wynne, and Mark Lavoie (2012). Monetary Economics. Palgrave Macmillan.

○ Empirical Studies

・Byrialsen, M.R., and Hamid Raza (2020). “An Empirical Stock-Flow Consistent Macroeconomic Model for Denmark.” Levy Economics Institute of Bard College Working Paper, 942.

・Burgess, S., Oliver Burrows, Antoine Godin, Stephen Kinsella, and Stephen Millard (2016). “A Dynamic Model of Financial Balances for the United Kingdom.” Bank of England.

○ Research in Japan

・Ono, T., and Nishi, H. (2011). “Reconstructing Kaleckian model: Stock-Flow Consistent model.”

This paper introduced the SFC model for the first time in Japan, focusing on theoretical analysis without connecting to real data.

5. What is an SFC Model?

○ A macroeconomic dynamic model that divides economic agents into households, firms, government, financial institutions, and the foreign sector, and models transactions within and between these sectors based on the principles of accounting entries.

○ It utilizes a framework known as the Transaction Flow Matrix.

		Households	Firms		Government	Banks		Rest of the world	Σ
			current	capital		current	capital		
Flow	Consumption	-C	+C						0
	Investment		+I	-I					0
	Export		+EX					-EX	0
	Import		-IM					+IM	0
	Government Expenditure		+G		-G				0
	Wage	+WB_h	-WB					+WB_o	0
	Entrepreneual profits	+FD_f	-F_f	+FU_f					0
	Bank profit	+FD_b				-F_b	+FU_b		0
	Loan interest	-$rl.L_{h-1}$	-$rl.L_{f-1}$			+$rl.L_{-1}$		-$rl.L_{o-1}$	0
	Deposit intetrest	+$rm.M_{h-1}$	+$rm.M_{f-1}$			-$rm.M_{-1}$		+$rm.M_{o-1}$	0
	Government bonds interest				-$rb.B_{-1}$	+$rb.B_{-1}$			0
	Tax	-T_h	-T_f		+T	-T_b			0
Stock	Loan	+△L_h		+△L_f			-△L	+△L_o	0
	Deposit	-△M_h		-△M_f			+△M	-△M_o	0
	Government bonds				+△B		-△B		0
	Equity	-△$ef.pef$		+△$ef.pef$					0
	Σ	0	0	0	0	0	0	0	0

○Horizontal and vertical totals sum to zero(Principle of accounting balance)
- Horizontal: Transactions between sectors
- Vertical: Sectoral internal balances

○Equations are set up within the constraints of these accounting relationships.

○Simulation results are output as being equivalent to the results of accounting entries.

6. Transaction Flow Matrix of the Japanese Economy <GDP and NDP>

< 2019 National Accounts >

(*1)

Fiscal Year (Billion Yen)	Households		Non_financial firms		Government		Financial firms		Non_profit inst.		Rest of the world		Σ
	current	capital	current	capital	current	capital	current	capital	current	capital			
1. Private final consumption expenditure	-296,451.0		303,858.8						-7,407.8				0.0
2. Government final consumption expenditure			111,826.9		-111,826.9								0.0
(including)													
(1) Individual consumption expenditure			68,703.1										
(2) Collective consumption expenditure			43,123.9										
3. Private fixed capital formation		-21,121.3	121,131.2	-94,960.4				-2,533.9		-2,515.6			0.0
(including) Changes in inventories			878.3										
4. Government fixed capital formation			22,078.9			-22,078.9							0.0
5. Net exports of goods and services			-2,059.5								2,059.5		0.0
<Gross Domestic Product>			556,836.3										
1. Private comsumption of fixed capital		23,820.1	-115,733.5	86,840.9				2,542.3		2,530.2			0.0
2. Government comsumption of fixed capital			-19,029.0			19,029.0							0.0
<Net Domestic Product>			422,073.8										

○To use data prior to the COVID-19 pandemic as the initial values for the simulation, data from the 2019 National Accounts have been used.

(*1) The horizontal totals represent transactions between sectors and sum to zero (though there may be discrepancies due to rounding).

6. Transaction Flow Matrix of the Japanese Economy ＜Primary Income Balance＞

＜ 2019 National Accounts ＞

(*1)

Fiscal Year (Billion Yen)		Households		Non_financial firms		Government		Financial firms		Non_profit inst.		Rest of the world		Σ
		current	capital	current	capital	current	capital	current	capital	current	capital			
<Net Domestic Product>				422,073.8										
1. Taxes on production and imports less Subsidies				-43,305.9		43,305.9								0.0
2. Operating surplus and mixed income		32,394.3		50,425.6	-92,292.3			9,472.4						0.0
3. Compensation of employees		287,994.7		-287,887.9								-106.8		0.0
(including)														
(1) Wages and salaries		244,185.4												
(2) Employers' social contributions		43,809.3												
a. Employers' actual social contributions		42,091.8												
b. Employers' imputed social contributions		1,717.5												
4. Property income		21,215.1		-3,117.0		37.7		3,391.1		274.2		-21,800.8		0.3
(1) Interest	(Receivable)	7,162.3		6,470.4		5,688.6		28,578.1		136.4		4,055.3		0.0
	(Payable)	-1,801.1		-3,049.6		-8,175.8		-24,593.5		-33.7		-14,437.4		
(2) Dividends	(Receivable)	6,383.5		19,004.5		2,524.5		16,520.3		170.0		8,308.2		0.2
	(Payable)	0.0		-25,596.0		0.0		-7,587.9		0.0		-19,726.9		
(3) Other investment income	(Receivable)	9,470.4		53.7		0.4		4.4		1.5				0.1
	(Payable)	0.0		0.0		0.0		-9,530.3		0.0				
(including) Investment income on pension entitlements	(Receivable)	1,257.4												0.0
	(Payable)							-1,257.4						
5. Rent		2,902.6		-2,524.5		-348.2		-124.4		33.2		61.3		0.0
	(Receivable)	3,149.4		1,642.8		19.5				54.8		78.7		0.0
	(Payable)	-246.8		-4,167.3		-367.7		-124.4		-21.6		-17.4		
<Balance of primary incomes, net>		344,506.7		43,371.8		42,995.4		12,739.1		307.4		-19,786.8		

6. Transaction Flow Matrix of the Japanese Economy < Disposable Income >

< 2019 National Accounts >

(*1)

Fiscal Year (Billion Yen)	Households		Non_financial firms		Government		Financial firms		Non_profit inst.		Rest of the world		Σ
	current	capital	current	capital	current	capital	current	capital	current	capital			
<Balance of primary incomes, net>	344,506.7		43,371.8		42,995.4		12,739.1		307.4		-19,786.8		
1. Current taxes on income, wealth, etc.	-30,356.4		-21,308.3		56,424.7		-4,760.0						0.0
2. Net social contributions	-83,912.5		947.9		74,398.5		8,416.3		149.8		0.0		0.0
(1) Employers' actual social contributions	-42,091.8				33,478.9		8,612.9						0.0
(2) Employers' imputed social contributions	-1,717.5		947.9		2,352.7		-1,732.9		149.8				0.0
(3) Households' actual social contributions	-39,138.6				38,566.9		571.7						0.0
(4) Households' contributions supplements	-1,257.4						1,257.4						0.0
(less) Service charges on pension scheme	292.8						-292.8						0.0
3. Social benefits other than social transfers in kind	79,518.1		-947.9		-68,842.6		-9,213.1		-514.6		0.0		-0.1
(1) Social security benefits in cash	59,054.3				-59,054.3								0.0
(2) Other social insurance pension benefits	9,166.0						-9,166.0						0.0
(3) Other social insurance non-pension benefits	3,497.4		-947.9		-2,352.7		-47.1		-149.8				-0.1
(4) Social assistance benefits	7,800.4				-7,435.6				-364.8				0.0
4. Other current transfers	-1,474.6		-2,170.8		-6,531.2		-88.9		9,107.2		1,158.5		0.2
(1) Net non-life insurance premiums (Receivable)							5,261.2						0.0
(Payable)	-3,102.5		-1,930.4		-18.3		-135.9		-74.1				
(2) Non-life insurance claims (Receivable)	3,030.4		1,862.3		14.4		289.5		64.7				0.1
(Payable)							-5,261.2						
(3) Current transfers within general governmer (Receivable)					66,777.6								0.0
(Payable)					-66,777.6								
(4) Current international cooperation (Receivable)					0.5								
(Payable)					-362.6								
(5) Miscellaneous current transfers (Receivable)	12,589.7		3,075.6		2,202.6		551.6		9,116.6				
(Payable)	-13,992.2		-5,178.3		-8,367.8		-794.1						0.1
(6) Other current transfers (Rest of the world) (Receivable)											686.9		
To/From Government sector (Payable)											-485.1		
(7) Other current transfers (Rest of the world) (Receivable)											5,002.7		
To/From Government other sectors (Payable)											-4,046.0		
<Disposable income, net>	308,281.3		19,892.7		98,444.8		7,093.4		9,049.8		-18,628.3		

6. Transaction Flow Matrix of the Japanese Economy <Net Lending / Net Borrowing>

< 2019 National Accounts >

(*1)

Fiscal Year (Billion Yen)	Households		Non_financial firms		Government		Financial firms		Non_profit inst.		Rest of the world		Σ
	current	capital	current	capital	current	capital	current	capital	current	capital			
<Disposable income, net>	308,281.3		19,892.7		98,444.8		7,093.4		9,049.8		-18,628.3		
Adjustment for the change in pension entitlements	-796.7						796.7						0.0
<Saving, net> (Disposable income, net - Consumption ± Adjustment for the change in pension entitlements)	11,033.6		19,892.7		-13,382.1		7,890.1		1,642.0		-18,628.3		
Savings investment balance (Saving, net - fixed capital formation + comsumption of fixed capital)	13,732.4		11,773.2		-16,432.0		7,898.5		1,656.6		-18,628.3		0.4
Capital transfers, etc.	2,136.6		-396.8		-907.6		-1,155.3		-137.3		460.4		0.0
(1) Capital transfers, net	-2,030.4		2,581.9		159.9		-1,191.9		20.1		460.4		0.0
(2) Changes in inventories	12.5		902.0	-878.3	-36.2								0.0
(3) Purchases of land, net	-4,179.5		2,955.0		1,103.7		-36.6		157.4				0.0
(*2) ① **<Net lending(+)/net borrowing(-)>**	15,869.0		11,376.4		-17,339.6		6,743.2		1,519.3		-18,167.9		0.4
<Changes in financial assets and liabilities>	Assets	Liabilities	Assets	Liabilities	Assets	Liabilities	Assets	Liabilities	Assets	Liabilities	Assets	Liabilities	
Monetary gold and SDRs	0.0	0.0	0.0	0.0	61.3	0.0	0.0	0.0	0.0	0.0	0.0	61.3	0.0
Currency and deposits	21,004.1	0.0	8,366.5	0.0	-2,808.0	0.0	251.6	27,155.2	649.9	0.0	-628.2	-319.3	0.0
Loans	26.5	7,854.5	1,258.1	13,236.5	-4,630.8	-2,199.7	121,070.9	98,420.4	24.5	221.9	28,712.0	28,927.6	0.0
Debt securities	1,683.8	0.0	1,200.1	10,079.0	-4,027.6	13,782.0	25,939.8	4,492.7	-1,037.4	0.0	4,595.0	0.0	0.0
Equity and investment fund shares	-2,822.4	0.0	-107.9	278.9	590.6	-1.4	12,529.2	11,536.9	152.4	0.0	1,472.5	0.0	0.0
Insurance, pension and standardized guarantee schemes	1,020.7	0.0	89.2	-1,374.2	0.0	0.0	-1,664.1	820.0	0.0	0.0	0.0	0.0	0.0
Financial derivatives and employee stock options	16.0	0.0	0.0	16.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other financial assets	774.4	-1,076.4	2,145.1	-20,833.2	10,965.2	1,836.8	18,693.0	29,641.3	-220.7	-861.4	25,339.5	48,989.4	0.0
(*2) ② **<Net lending(+)/net borrowing(-) (financial surplus or deficit)>**		14,925.0		11,548.1		-13,267.0		4,753.9		208.2		-18,168.2	0.0
① - ② (statistical error)		944.0		-171.7		-4,072.6		1,989.3		1,311.1		0.3	0.4

(*2) Although ① and ② are supposed to match, statistical errors arise because ① is an aggregate from the capital account and ② is an aggregate from the financial account.

7. Features of the Model

○ Interest Rates

・SNA's asset and liability items are classified into six types: interest-bearing assets and liabilities, equity assets and liabilities, and investment assets and liabilities.

⇨ Return rates are set for each sector and each type (6 sectors × 6 types = 36 rates).

Firms	① Asset	② Composition ratio (①÷⑤)	③ Return rate (④÷①)	④ Asset income (receivable)	① Liability	② Composition ratio (①÷⑤)	③ Return rate (④÷①)	④ Asset income (payable)
A Interest bearing	664,378	0.58	0.97%	6,470	843,158	0.50	0.36%	3,050
B Equity	470,511	0.41	4.04%	19,005	819,751	0.48	3.12%	25,596
C Investment	3,263	0.00	1.65%	54	27,578	0.02	0.00%	0
⑤ Total	1,138,152				1,690,487			

A Interesting bearing : Currency and deposits, Loans, Debt securities, Financial derivatives and employee stock options, Other financial assets

B Equity : Equity and investment fund shares

C Investment : Insurance, pension and standardized guarantee schemes

7. Features of the Model

○ Wages

・Wages are assumed to grow at a fixed rate of increase, while decreasing by the rate of population decline (1%).

Domestic wage WB_d = (1 + grwb)・WB_d(-1)・(1 + grpp)

grwb：Wage increase rate、grpp：Population growth rate

○ Consumption

・Consumption is modeled as a function of expected disposable income and net financial assets.

・Expected disposable income is assumed to grow at the same rate as wages.

Households consumption C_h = α1・Yde + α2・V_h(-1)

α1：Propensity to consume out of disposal income、α2：Propensity to consume out of net financial asset, V_h：Households net financial asset

Expected disposable income Yde = (1 + grwb)・YD(-1)

7. Features of the Model

○ Corporate Investment

・ A function of the constant term (C), capacity utilization (GDP/FixCapital), interest rates (ShortRate), and GDP

Firms Investment I_nf ＝ - 100673.4 - 40773.64・GDP/Fixed Capital + 367034.6・Shortloan Rate + 0.405041・GDP

1994~2019	Coefficient				AIC	Adj R^2
	C	GDP	GDP/FixCapital	ShortRate		
GDP/FixCapital, ShortRate			113884.3**	-495452.8**	20.30279	0.166312
C, GDP/FixCapital, ShortRate	7471.049		104023.4*	-455864.6	20.37743	0.132046
GDP		0.162123**			19.55629	0.590286
C, GDP	-98413.53**	0.348079**			18.77463	0.819144
ShortRate				4700210**	24.00752	-34.125634
C, ShortRate	85730.11**			-7802.736	20.52543	-0.041588
GDP/FixCapital			104799**		20.50824	-0.061466
C, GDP/FixCapital	48569.70		45412.63		20.43555	0.047944
C, GDP/FixCapital, ShortRate, GDP	**-100673.4****	**0.405041****	**-40773.64**	**367034.6****	**18.55232**	**0.864556**

・ The coefficient for interest rates is positive: An increase in interest rates boosts GDP through income effects, which in turn drives investment.

・ The coefficient for capacity utilization is negative: Due to uncertainty about the future, domestic investment does not increase as much as GDP growth.

7. Features of the Model

○ Government Expenditure

・Public investment grows at a constant rate.

Government investment I_g = (1 + grg)・G(-1)

grg：Growth rate of government investment

・Government collective consumption and government individual consumption (in-kind benefits) are expressed as a ratio to GDP

Government individual consumption C_g_ind = μind・Y

μind：Government individual consumption/GDP

Government collective consumption C_g_col = μcol・Y

μcol：Government collective consumption/GDP

7. Features of the Model

○ Cash transfers and social contributions

・Cash transfers and social contributions are expressed as a ratio to GDP and allocated to each sector based on current ratios.

Households social benefit(Receivable) SBEN_h = ωbh・Y

ωbh：Households social benefit/GDP

(Payable) ⇨ Firms : 0.012, Government : 0.864, Financial firms : 0.116, Non_profit inst. : 0.008

Households social contributions(Payable) SCON_h = ωch・Y

ωch：Households social contributions/GDP

(Receivable) ⇨ Firms : 0.011, Government : 0.887, Financial firms : 0.100, Non_profit inst. : 0.002

7. Features of the Model

○ Net Exports

・ Expressed as a ratio to GDP.

Net exports NEX = ε ・Y

ε ：Net exports/GDP

○ Exogenous parameters such as ratios to GDP, growth rate are calibrated using data from 1994 to 2019. (Adjustments are made as needed based on changes in trends or other circumstances.)

8. The model equations

＜Output＞		
Gross Domestic Product	Y = C_p + I_p + G + NEX	C_p：Private consumption、I_p：Private investment、G：Government expenditure、NEX：Net exports
Net Domestic Product	NY = (1 − δ) · Y	δ：Consumption of fixed capital/GDP
＜Consumption＞		
Private consumption	C_p = C_h + C_npi	
Households consumption	C_h = α1 · Yde + α2 · V_h(-1)	α1：Propensity to consume out of disposal income、α2：Propensity to consume out of net financial asset、V_h：Households net financial asset
Non_profit inst. consumption	C_npi = ι · Y	τ：Non_profit inst. consumption/GDP
＜Investment＞		
Private investment	I_p = I_h + I_nf + I_f + I_npi	
Households investment	I_h = βh · Y-1(-1)	βh：Households investment/GDP
Firms investment	I_nf = - 100673.4 - 40773.64*GDP/Fixed Capital + 367034.6*Shortloan Rate + 0.405041*GDP	
Financial firms investment	I_f = βf · Y(-1)	βf：Financial firms investment/GDP
Non_profit inst. investment	I_npi = βnpi · Y(-1)	βnpi：Non_profit inst. investment/GDP
＜Government expenditure＞		
Government expenditure	G = C_g + I_g	
Government consumption	C_g = C_g_ind + C_g_col	
Government individual consumption	C_g_ind = μind · Y	μind：Government individual consumption/GDP
Government collective consumption	C_g_col = μcol · Y	μcol：Government collective consumption/GDP
Government investment	I_g = (1 + grg) · G(-1)	grg：Growth rate of government investment
＜Net exports＞		
Net exports	NEX = ε · Y	ε：Net exports/GDP
＜Consumption of fixed capital＞		
Total consumption of fixed capital/GDP	δ = δh + δnf + δg + δf + δnpi	δh：Households consumption of fixed capital/GDP、δnf：Firms consumption of fixed capital/GDP、δg：Government consumption of fixed capital/GDP、δf：Financial firms consumption of fixed capital/GDP、δnpi：Non_profit inst.consumption of fixed capital/GDP
＜Tax＞		
Total tax	T = (θi + θdh + θdnf + θdf) · Y	θi：Indirect tax/GDP、θdh：Households direct tax/GDP、θdnf：Firms direct tax/GDP、θdf：Financial firms direct tax/GDP

8. The model equations

＜Operating surplus＞		
Total operating surplus	OS = χ ・Y	χ ：Total operating surplus/GDP
Households operating surplus	OS_h = χh ・OS	χh：Households operating surplus/Total operating surplus
Firms operating surplus	OS_nf = χnf ・OS	χnf：Firms operating surplus/Total operating surplus
Financial firms operating surplus	OS_f = χf ・OS	χf：Financial firms operating surplus/Total operating surplus
＜Wage＞		
Total wage	WB = WB_d + Wb_o	
Domestic wage	WB_d = (1 + grwb) ・WB_d(-1) ・(1 + grpr + grpp)	grwb：Wage increase rate、grpp：Population growth rate
Rest of the world wage	WB_o = σ ・Y	σ ：Rest of the world wage/GDP
＜Property income＞		
Sectoral(*)Property income * = h, nf, g, npi, o	PI_* = V_IBA_* ・R_IBA_* + V_EQA_* ・R_EQA_* + V_ASA_* ・R_ASA_* + V_IBL_* ・R_IBL_* + V_EQL_* ・R_EQL_* + V_ASL_* ・R_ASL_*	V_IBA_*：Interest bearing asset、R_IBA_*：Yield of interest bearing asset、 V_EQA_*：Equity asset、R_EQA_*：Yield of equity asset、 V_ASA_*：Asset under management、R_ASA_*：Yield of asset under management、 V_IBL_*：Interest bearing liability、R_IBL_*：Yield of interest bearing liability、 V_EQL_*：Equity liability、R_EQL_*：Yield of equity liability、 V_ASL_*：Liability of asset under manegemant、R_ASL_*：Yield of liability of asset under management
Financial firms property income	PI_f = -(PI_h + PI_nf + PI_g + PI_npi + PI_o)	PI_h：Households property income、PI_nf：Firms property income、PI_g：Government property income、 PI_npi：Non_profit inst. property income、PI_o：Rest of the worlds property income
＜Rent＞		
Households rent/GDP	κh = -(κnf + κg + κf + κnpi + κo)	κnf：Firms rent、κg：Government rent、κf：Financial firms rent、κnpi：Non_profit inst. rent、 κo：Rest of the world rent
＜Social benefit other than social transfers in kind＞		
Households social benefit(Receivable)	SBEN_h = ωbh ・Y	ωbh：Households social benefit/GDP
Firms social benefit(Payable)	SBEN_nf = SBEN_h ・0.012	
Government social benefit(Payable)	SBEN_g = SBEN_h ・0.864	
Financial firms social benefit(Payable)	SBEN_f = SBEN_h ・0.116	
Non_profit inst. social benefit(Payable)	SBEN_npi = SBEN_h ・0.008	

8. The model equations

<Social contributions>		
Households social contributions(Payable)	SCON_h = ωch・Y	ωch：Households social contributions/GDP
Firms social contributions(Receivable)	SCON_nf = SCON_h・0.011	
Government social contributions(Receivable)	SCON_g = SCON_h・0.887	
Financial firms social contributions(Receivable)	SCON_f = SCON_h・0.100	
Non_profit inst. social contributions(Receivable)	SCON_npi = SCON_h・0.002	
<Other current transfers>		
Households other current transfers/GDP	ρh = -(ρnf + ρg + ρf + ρnpi + ρo)	ρnf：Firms other current transfers/GDP、ρg：Government other current transfers/GDP、 ρf：Financial firms other current transfers/GDP、ρnpi：Non_profit inst. other current transfers/GDP、 ρo：Rest of the world other current transfers/GDP、
<Capital transfers, etc.>		
Households capital transfers, etc./GDP	λh = -(λnf + λg + λf + λnpi + λo)	λnf：Firms capital transfers, etc./GDP、λg：Government capital transfers, etc./GDP、 λf：Financial firms capital transfers, etc./GDP、λnpi：Non_profit inst. capital transfers, etc./GDP、 λo：Rest of the world capital transfers, etc./GDP、
<Households>		
Disposable income	YD = WB_d + OS_h + PI_h + κh・Y - θdh・Y - SCON_h + SBEN_h + ρh・Y	
Expected disposable income	Yde = (1 + grwb)・YD(-1)	
Adjustment for the change in pension entitlements	PEN_adj_h = SCON_h - SBEN_h	
Net lending	NL_h = YD - C_h + PEN_adj_h - I_h + δh・Y + λh・Y	
Net financial asset	V_h = V_IBA_h + V_EQA_h + V_ASA_h + V_IBL_h + V_EQL_h + V_ASL_h	
<Firms>		
Entrepreneurial profits	F = NY - θi - OS + OS_nf - WB + PI_nf + κnf・Y	
Retained earnings	FU = F - θdnf・Y + SCON_nf - SBEN_nf + ρnf・Y	
Net lending	NL_nf = FU - I_nf + δnf・Y + λnf・Y	
Net financial asset	V_nf = V_IBA_nf + V_EQA_nf + V_ASA_nf + V_IBL_nf + V_EQL_nf + V_ASL_nf	
Net financial asset(Issued share adjusted)	V_nf_adj = V_nf - V_EQL_nf	

8. The model equations

<Government>		
Net lending	NL_g = T - G + PI_g + κg・Y + SCON_g - SBEN_g + ρg・Y + δg・Y + λg・Y	
Net financial asset	V_g = V_IBA_g + V_EQA_g + V_ASA_g + V_IBL_g + V_EQL_g + V_ASL_g	
Government debt	DEBT_g = -(V_IBL_g + V_EQL_g + V_ASL_g)	
<Financial firms>		
Adjustment for the change in pension entitlements	PEN_adj_f = -PEN_adj_h	
Net lending	NL_f = OS_f + PI_f + κf・Y - θdf・Y + SCON_f - SBEN_f + ρf・Y + PEN_adj_f - I_f + δf・Y + λf・Y	
Net financial asset	V_f = V_IBA_f + V_EQA_f + V_ASA_f + V_IBL_f + V_EQL_f + V_ASL_f	
<Non_profit inst.>		
Net lending	NL_npi = PI_npi + κnpi・Y + SCON_npi - SBEN_npi + ρnpi・Y - C_npi - I_npi + δnpi・Y + λnpi・Y	
Net financial asset	V_npi = V_IBA_npi + V_EQA_npi + V_ASA_npi + V_IBL_npi + V_EQL_npi + V_ASL_npi	
<Rest of the world>		
Net lending	NL_o = -NEX + WB_o + PI_o + κo・Y + ρo・Y + λo・Y	
Net financial asset	V_o = V_IBA_o + V_EQA_o + V_ASA_o + V_IBL_o + V_EQL_o + V_ASL_o	

9. Simulation Results

	Parameters								Landing value after 20 years				
Verification period : 2025~2045	Public investment (Growth rate)	Cash transfers (Ratio to GDP)	In-kind benefits (Ratio to GDP)	Social contribtuions (Ratio to GDP)	Wage (Growth rate)	Propensity to consume out of YD	Propensity to consume out of n.f.asset	Change in interest rate	GDP growth rate	#Government debt (Ratio to GDP)	Firms net financial asset (Multiplier)	Households net financial asset (Multiplier)	Change in tax revenue (Multiplier)
①Baseline	1.28%	14.28%	12.34%	15.07%	0.39%	0.772	0.045		0.29%	313%	**8.04**	1.27	1.08
②Interest rate hike	1.28%	14.28%	12.34%	15.07%	0.39%	0.772	0.045	**+1%**	1.13%	347%	**12.44**	1.56	1.32
③Wage increase	1.28%	14.28%	12.34%	15.07%	**3.39%**	0.772	0.045		**2.85%**	**164%**	**6.53**	1.56	2.02
④Increase in Cash Transfers	1.28%	**15.53%**	12.34%	15.07%	0.39%	0.772	0.045		**0.44%**	319%	**8.46**	1.28	1.12
⑤Increase in In-Kind Benefits + Wage Increase	1.28%	14.28%	**13.59%**	15.07%	**3.39%**	0.772	0.045		**3.01%**	**172%**	**7.27**	1.56	2.10

#The government debt-to-GDP ratio for 2019 is 238%.

① Baseline

② Interest Rate Increase: Assume an increase of 1%

③ Wage Increase: Based on recent policy trends and recent performance, assume a continuation of a 3% wage increase rate in the future

④ Increase in Cash Transfers: Assume an expenditure equivalent to 1% of the increase in interest rates (1% of government net debt) ⇨ Comparison of the effects of an interest rate hike and cash transfers

⑤ Increase in In-Kind Benefits + Wage Increase: Assume an expenditure equivalent to 1% of the increase in interest rates (1% of government net debt) + 3% wage increase rate ⇨ Evaluation of the effects of implementing JGP

10. Analysis of Simulation Results

① Baseline

・A tendency for firms to accumulate internal reserves can be observed.

・Firms have been investing these internal reserves abroad rather than in the domestic market, where growth prospects are limited.

⇨ Even if corporate performance is strong, there has been no tangible sense of it.

② Interest Rate Increase

・Generally, increasing interest rates are thought to suppress growth by reducing investment; nonetheless, in Japan, an increase in interest rates tends to stimulate growth through income effects.

・ However, much of the income remains as internal reserves within companies, and its contribution to growth through increased consumption is limited.

③ Wage Increase

・Wage increases significantly boost the economy through expanded consumption.

・The transfer of income from firms to households leads to a reduction in corporate internal reserves.

・Increased tax revenues from economic growth substantially reduce the government debt-to-GDP ratio.

10. Analysis of Simulation Results

④ Increase in Cash Transfers

- While there is a growth-stimulating effect, corporate internal reserves increase.
- The transfer of income from firms to households does not advance significantly, resulting in a limited contribution to growth through expanded consumption.

⑤ Increase in In-Kind Benefits + Wage Increases

- The growth-stimulating effect of increased in-kind benefits is enhanced by the effect of wage increases, leading to significant economic growth.
- The increase in tax revenues resulting from this growth causes a substantial reduction in the government debt-to-GDP ratio.

11. Summary: Insights for Revitalizing the Japanese Economy

○ Wage increase

Income Transfer from Firms to Households through Wage Increases is Key.
However, in Japan, where labor unions are weak, achieving wage increases presents a significant challenge.

○ Interest Rate Increase

Contrary to conventional wisdom, raising interest rates can stimulate the economy through income effects (consistent with MMT claims).
However, there are potential drawbacks:

- Benefits primarily accrue to asset holders, with minimal impact on overall consumption.
- Without accompanying wage increases, corporate internal reserves are likely to rise without boosting consumption.

11. Summary: Insights for Revitalizing the Japanese Economy

○ Increase in Cash Transfers

Generally considered to have high redistributive effects, but not as effective as wage increases.

How to Achieve Wage Increases?

○ Implementing a Job Guarantee Program

・This program could establish a real wage floor, potentially leading to higher overall wage levels.

・An increase in wage levels → Increase in consumption → Increase in investment → Increase in income → Increase in consumption…

⇨ This cycle is expected to emerge, leading to a way out of the deflationary spiral.